

of the Patriot.

So what have we learned from this? Well, the error could have been reduced if the embedded system was capable of handling more than 24 bits, however it would be better to remove rather than just reduce. So the relative time between two radar pulses was being measured. In addition to this the clock should always be reset at the first pulse and stopped at the second pulse. This avoids the accumulation of errors and so also avoids the subtraction of two huge identical numbers.

Pathfinder

NASA launched their space exploration robot, the Mars Pathfinder on December 4, 1996. It was supposed to be a 'faster, better and cheaper' spacecraft, but unfortunately that was not to be. It landed successfully on Mars on July 4, 1997 and then started the collection of data like soil samples, sending images back to earth, and conducting chemical analysis. A few days later though, things had changed. The robot would frequently experience a total system reset because of which it could not function normally and so would always fall to serious data loss.

Pathfinder fell victim to a software system failure called Priority inversion. Pathfinder here was using VxWorks, a real time embedded operating system. This system is responsible to schedule all the tasks and management of all the system resources, for example memory. Using this system, tasks are scheduled in such a manner that one task is never executed continuously, but is instead divided into fragments and interweaved. The tasks are all assigned an appropriate priority. At one time slot the system will perform task 1, and for the next time slot if there is a task of higher priority (say task 2), the system will execute task 2. If there is no task of higher priority than task 1 then task 1 will continue.

For tasks to read and write on, a piece of shared resource called the 'information bus' is used in the same system. What happens here is that data stored by one task into a shared memory, if it has no protection mechanism, will always be overwritten or misread by tasks which are scheduled to perform later. So in simple words, if a task sees that the shared memory is used by another task, it will politely wait for the other task to finish using the shared memory with disregard to the priority levels.

The failure involved three tasks- the meteorological data gathering task (low priority), data transmission task (medium priority), and the information management task (high priority). Before the failure occurred,